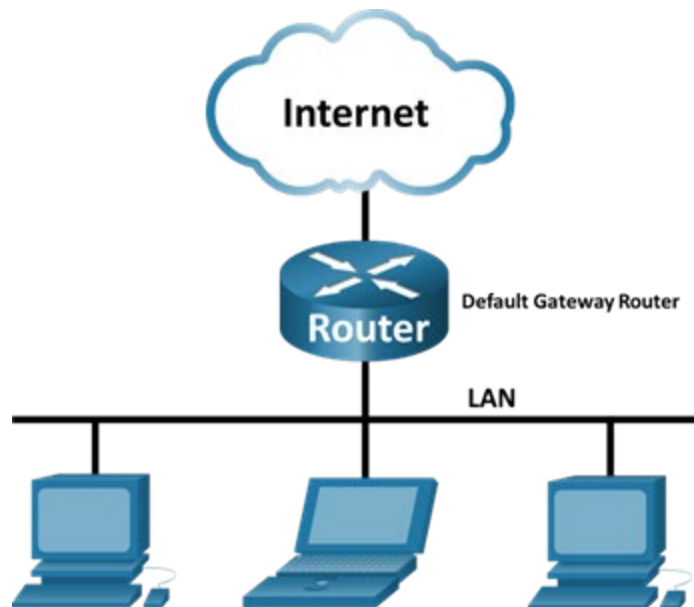


## Lab - View Captured Traffic in Wireshark in a Local Network

### Example Topology



### Objectives

#### Part 2: Capture and Analyze ARP Data in Wireshark

- Start and stop data capture of ping traffic to remote hosts.
- Locate the IPv4 and MAC address information in captured PDUs.
- Analyze the content of the ARP messages exchanged between devices on the WLAN/LAN.

#### Part 3: View the ARP cache entries on the PC

- Access the Windows Command Prompt.
- Use the Windows arp command to view the local ARP table cache on the PC.

### Background / Scenario

Address Resolution Protocol (ARP) is used by TCP/IP to map a Layer 3 IPv4 address to a Layer 2 MAC address. When an Ethernet frame is transmitted on the network, it must have a destination MAC address. To dynamically discover the MAC address of a known destination, the source device broadcasts an ARP request on the local network. The device that is configured with the destination IPv4 address responds to the request with an ARP reply and the MAC address is recorded in the ARP cache.

Every device on the LAN maintains its own ARP cache. The ARP cache is a small area in RAM that holds the ARP responses. Viewing an ARP cache on a PC displays the IPv4 address and the MAC address of each device on the LAN with which the PC has exchanged ARP messages.

Wireshark is a software protocol analyzer, or "packet sniffer" application, used for network troubleshooting, analysis, software and protocol development, and education. As data streams travel back and forth over the network, the sniffer "captures" each protocol data unit (PDU) and can decode and analyze its content according to the appropriate protocol specifications.

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Wireshark is a useful tool for anyone working with networks and can be used with most labs in the Cisco courses for data analysis and troubleshooting. This lab provides instructions for downloading and installing Wireshark, although it may already be installed. In this lab, you will use Wireshark to capture ARP exchanges on the local network.

## Part 2: Capture and Analyze Local ARP Data in Wireshark

In Part 2 of this lab, you will ping another PC on the WLAN/LAN and capture ARP requests and replies in Wireshark. You will also look inside the frames captured for specific information. This analysis should help to clarify how packet headers are used to transport data to their destination.

### Step 1: Retrieve your PC's interface addresses.

For this lab, you will need to retrieve your PC's IPv4 address and the MAC address.

- a. Navigate to a Command Prompt window, type **ipconfig /all** at the prompt.
- b. Note which network adapter that the PC is using to access the network. Record your PC interface's IPv4 address and MAC address (Physical Address).

```
C:\Users\Student> ipconfig /all
```

```
Wireless LAN adapter Wireless Network Connection:
```

```
Connection-specific DNS Suffix  :
Description . . . . . : Intel(R) Centrino(R) Advanced-N 6205
Physical Address. . . . . : A4-AE-31-AD-78-4C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . . :80::f9e7:e41d:a772:f993%11(Preferred)
IPv4 Address. . . . . : 192.168.1.8(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, August 04, 2016 05:35:35 PM
Lease Expires . . . . . : Friday, August 05, 2016 05:35:35 PM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 245648945
DHCPv6 Client DUID. . . . . : 00-01-00-01-1B-87-BF-52-A4-4E-31-AD-78-4C
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Disabled
```

- c. Ask a team member for their PC's IPv4 address and give your PC's IPv4 address to them. Do not provide them with your MAC address at this time.

**Question:** Write the IPv4 addresses of the default gateway and the other PC on the LAN.

**Answer IPv4:**

### Step 2: Start Wireshark and begin capturing data.

- a. On your PC, click **Start** and type **Wireshark**. Click **Wireshark Desktop App** when it appears in the search results window.
  - b. After Wireshark starts, select the network interface that you identified with the **ipconfig** command. Enter **eth.addr == AA:BB:CC:DD:EE:FF && arp** in the filter box. Please replace the MAC address AA:BB:CC:DD:EE:FF with your own MAC address. Note that Wireshark separates the MAC-address with colons.
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This selection configures Wireshark to only display packets that are part of your PC's ARP exchanges between the devices on the local network. After you have selected the correct interface and entered the filter information, click **Start capturing packets** (shark fin icon) to begin the data capture.

c. In a Command Prompt window, ping the default gateway of your network to test the connectivity to the default gateway address that was identified in Part 2, Step 1.

```
C:\Users\Student> ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=7ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=6ms TTL=64
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms
```

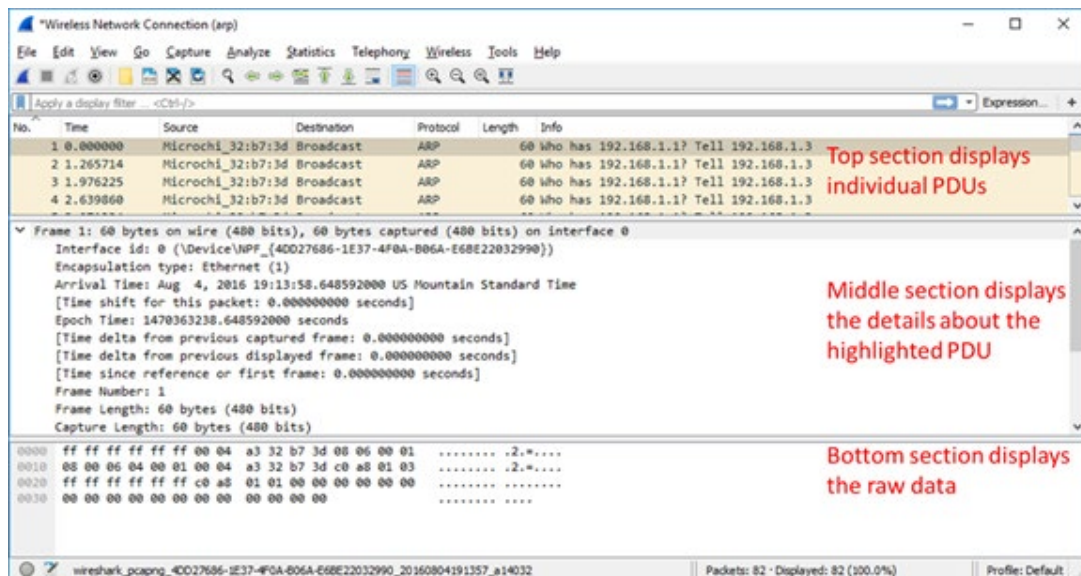
d. Ping the IPv4 addresses of other PCs on the LAN that were provided to you by your team members. **Note:** If your team member's PC does not reply to your pings, this may be because their PC firewall is blocking these requests. Ask your instructor for assistance to disable the PC firewall if necessary.

e. Stop capturing data by clicking **Stop Capture** (red square icon) on the toolbar.

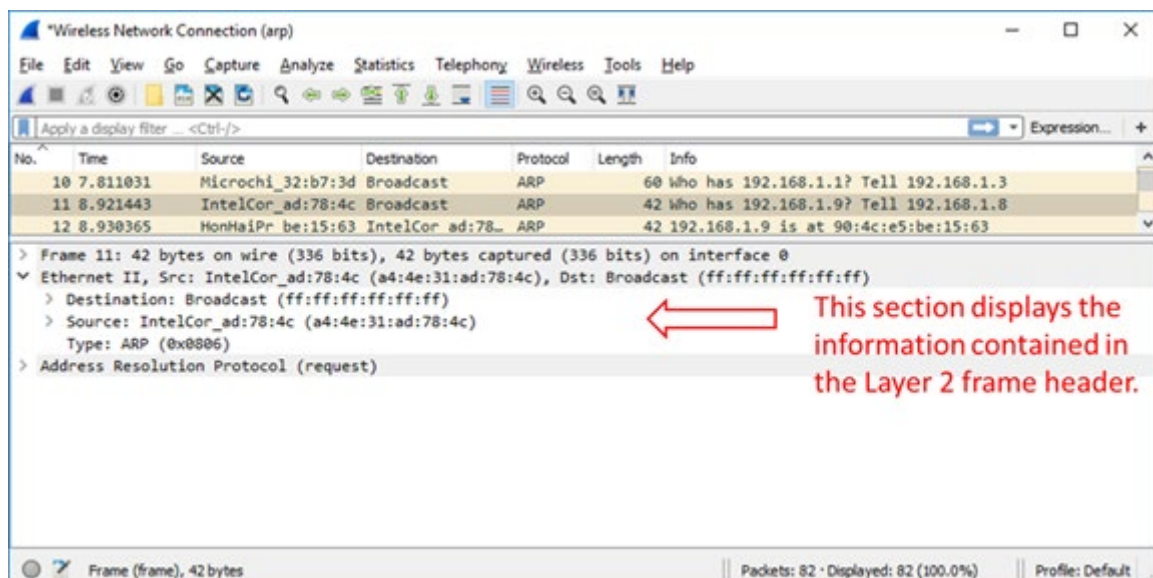
### Step 3: Examine the captured data.

In Step 3, examine the data that was generated by the **ping** requests of your team member's PC. Wireshark data is displayed in three sections:

- 1) The top section displays the list of PDU frames captured with a summary of the IPv4 packet information listed (see pictures next page).
  - 2) The middle section lists PDU information for the frame selected in the top part of the screen and separates a captured PDU frame by its protocol layers.
  - 3) The bottom section displays the raw data of each layer. The raw data is displayed in both hexadecimal and decimal form.
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- Click one of the ARP frames in the top section that has your PC MAC address as the source address in the frame and “broadcast” as the destination of the frame.
- With this PDU frame still selected in the top section, navigate to the middle section. Click the arrow to the left of the Ethernet II row to view the Destination and Source MAC addresses.

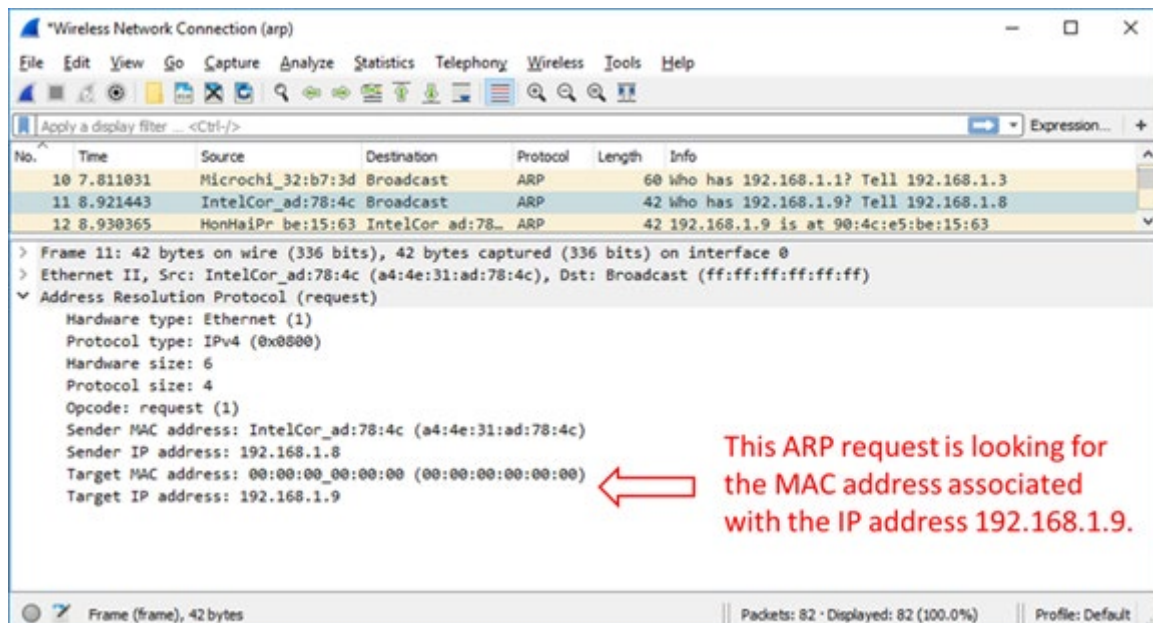


### Question:

Does the Source MAC address match your PC's interface?

### Answer:

- Click the arrow to the left of the Address Resolution Protocol (request) row to view the content of the ARP request.



**Step 4: Locate the ARP reply frame that corresponds to the ARP request that you highlighted.**

a. Using the Target IPv4 address in the ARP request, locate the ARP reply frame in the upper section of the Wireshark capture screen.

**Question:**

What is the IPv4 address of the Target device in your ARP request?

**Answer:**

b. Highlight the reply frame in the upper section of the Wireshark output. You may have to scroll the window to find the reply frame that matches the Target IPv4 address identified in the previous step. Expand the Ethernet II and Address Resolution Protocol (reply) rows in the middle section of the screen.

**Questions:**

Is the ARP reply frame a broadcast frame?

**Answer:**

What is the destination MAC address of the frame?

**Answer:**

Is this the MAC address of your PC?

**Answer:**

What MAC address is the source of the frame?

**Answer:**

c. Verify with your team member that the MAC address matches the MAC address of their PC.

### Part 3: Examine the ARP cache entries on the PC.

After the ARP reply is received by the PC, the MAC Address to IPv4 address association is stored in cache memory on the PC.

- a. Open a command prompt window on the PC. At the prompt, enter **arp -a** and press enter.

```
C:\Users\Student> arp -a
Interface: 192.168.1.8 --- 0xb
    Internet Address      Physical Address      Type
    192.168.1.1           00-37-73-ea-b1-7a     dynamic
    192.168.1.9           90-4c-e5-be-15-63     dynamic
    192.168.1.13          a4-4e-31-ad-78-4c     dynamic
    224.0.0.5             01-00-5e-00-00-05     static
    224.0.0.6             01-00-5e-00-00-06     static
    224.0.0.22            01-00-5e-00-00-16     static
    224.0.0.252           01-00-5e-00-00-fc     static
    224.0.0.253           01-00-5e-00-00-fd     static
    239.255.255.250       01-00-5e-7f-ff-fa     static
    255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

The output of the **arp -a** command displays the entries that are in the cache on the PC. In the example, the PC has entries for the default gateway (192.168.1.1) and for two PCs that are located on the same LAN (192.168.1.9 and 192.168.1.13).

#### Question:

What is the result of executing the **arp -a** command on your PC?

#### Answer:

- b. The **arp** command on the Windows PC has another functionality. Enter **arp /?** at the command prompt and press enter. The **arp** command options enable you to view, add and remove ARP table entries if necessary.

#### Question:

Which option deletes an entry from the ARP cache?

#### Answer:

#### Reflection

1. What is a benefit of keeping ARP cache entries in memory on the source computer?

#### Answer:

2. If the destination IPv4 address is not located on the same network as the source host, what MAC address will be used as the destination target MAC address in the frame?

#### Answer:

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